PARTS LIST

Item	Qty	REFERENCE DESIGNATOR	PART NUMBER	DESCRIPTION	MANUFACTURER
1	1		S1101EFF0300	BARE BOARD	ZIN TECHNOLOGIES
2	22	C1, C2, C3, C4, C5, C21, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C46, C47	C0805C104K1RACTU	CAPACITOR, CERAMIC, 0.1uF, 100V, 10%, 0805	KEMET
3	5	C22, C23, C24, C44, C45	T494B106K016AT	CAPACITOR 10.0uF 16V TANTALUM 10% B-SIZE SMD	KEMET
4	4	C25, C26, C27, C28	GA355DR7GF472KW01L	CAP, CER, SMT, 4700pF, C2220	MURATA
5	1	C29	CDR35BX224AKWR	CAPACITOR CERAMIC CHIP 0.22uF 50V ±10% MIL-PRF-55681 1825	KEMET
6	1	D1	1N4148TA	DIODE,TH,SWITCHING,100V,0.2A	FAIRCHILD
7	1	J13	SDD62M3S00G	CONN, SUB-D, 62 PIN, MALE, VERT	POSITRONIC
8	3	J14, J16, J18	SND15M3S00G	CONN, SUB-D, 15 PIN, MALE, VERT	POSITRONIC
9	2	J15, J17	SDD26M3S00G	CONNECTOR, SUB-D, 26 PIN, 3 ROW, MALE, VERTICAL WITH SWAGED SPACERS	POSITRONIC
10	3	Q1, Q2, Q3	2N7002	N-CHAN FET, SOT-23	FAIRCHILD
11	14	R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14	ERJ-8ENF4750V	RESISTOR 475 OHM 1/4W 1% 1206 SMD	PANASONIC - ECG
12	28	R16, R18, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78	ERJ-6ENF1002V	RESISTOR 10.0K 1/8W 1% 0805 SMD	PANASONIC - ECG
13	4	R17, R19, R48, R49	ERJ-6ENF1003V	RESISTOR, 100K, 1/8W, 1%, 0805	PANASONIC - ECG
14	12	R27, R28, R31, R34, R35, R36, R37, R38, R39, R40, R41, R42	ERJ-6ENF3321V	RESISTOR 3.320K 1/8W 1% 0805 SMD	PANASONIC - ECG
15	1	R47	ERJ-6ENF1001V	RESISTOR 1.00K 1/8W 1% 0805 SMD	PANASONIC - ECG
16	2	R79, R80	ERJ-6ENF22R1V	RESISTOR 22.1 OHM 1/8W 1% 0805 SMD	PANASONIC - ECG
17	1	R81	ERJ-6ENF3010V	RESISTOR, 301 OHM, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
18	1	R83	ERJ-6ENF75R0V	RESISTOR, 75 OHM, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
19	2	R84, R85	ERJ-6ENF37R4V	RESISTOR, 37.4 OHM, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
20	2	R86, R87	ERJ-8GEY0R00V	RESISTOR 0.0 OHM 1/4W 5% 1206 SMD	PANASONIC - ECG
21	1	U1	CD74HCT04M	IC, SMT, HEX INVERTERS, 14-SOIC	TEXAS INSTRUMENTS
22	1	U2	CD74HC32M	IC, QUAD 2-INPUT OR GATE, 14 PIN SOIC	TEXAS INSTRUMENTS
23	1	U3	CD74HC245M	IC BUS TRANSCEIVER 8BIT 20SOIC,	TEXAS INSTRUMENTS
24	7	U4, U5, U6, U7, U8, U9, U10	SZD20CF	RELAY, SMT, 2A, 60VDC, OPTO-ISOLATED SHORT-CIRCUIT PROTECTED	TELEDYNE
25	1	U11	V24C12C100BL2	MODULE, DC-DC CONVERTER, 12V, 100W, VICOR, -20°C TO +100°C	VICOR
26	1	U12	MAX4448ESE	SINGLE ENDED TO DIFFERENTIAL LINE DRIVER	MAXIM
27	A/R		TZ-211	LABEL, 1/4"BLK LETTERS WHT BACKROUND	BROTHER
28	A/R		3140	RTV FOR CONFORMAL COATING	DOW CORNING
29	A/R		PR-1204	PRIMER FOR CONFORMAL COATING	DOW CORNING
30	A/R		SN63PB37/RMA	SOLDER	
31	1		S1101MFA1526	60 DEGREE RADIOMETER ASSEMBLY	ZIN TECHNOLOGIES
32	4		NAS620C2	#2 FLAT WASHER REDUCED OUTER DIAMETER	
33	2		NAS1101E02-5	SCREW, MACHINE-FLAT FILLISTER HEAD .086-56 UNJC-3A X .312 LENGTH	
34	2		NAS1291C02M	NUT, .086-56 UNJC-3B, SELF LOCKING, A286	
35	6		JAX108-M09-F09-440	JACK SCREW, .112-40 UNC-2A X .625 LG THREAD, 303 STAINLESS STEEL, PASSIVATED	UNICORP INC.
36	6		JAX110-M09-F09-440	JACK SCREW, .112-40 UNC-2A X .750 LG THREAD, 303 STAINLESS STEEL, PASSIVATED	UNICORP INC.
37	36		NAS620C4	#4 FLAT WASHER REDUCED OUTER DIAMETER	
38	12		MS21043-04	#4-40 UNJC-3B REDUCED HEX SELF LOCKING REDUCED HEIGHT NUT	
39	2	R15, R82	ERJ-P08J511V	RESISTOR 510 OHM 2/3W 1% 1206 SMD	PANASONIC - ECG

8. CONFORMAL COATING, STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1 AND THE FOLLOWING NOTES:
   - A. PERFORM CLEANLINESS TEST. DO NOT SUBMERGE ITEM 31 INTO CLEANING SOLVENT.
   - B. BAKE PWA AT 93°C ±5.5°C (200°F ±10°F) FOR 4 HOURS PRIOR TO CONFORMAL COATING.
   - △C MASK OFF ALL DESIGNATED AREAS USING KAPTON TAPE. ALL MASKING OPERATIONS SHALL BE PERFORMED UNDER AN AIR IONIZER.
   - △D MASK OFF RADIOMETER HOUSING SUCH THAT RADIOMETER SENSOR WINDOW IS NOT COATED.
   - △E MEASURE CONFORMAL COATING THICKNESS AT DESIGNATED AREAS.
   - F. CONFORMAL COAT THE COMPLETE ASSEMBLY USING ITEM 28 AND ITEM 29.
9. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY AFTER CONFORMAL COATING PER TEST PLAN ACME-TST-026.
10. PLACE THE COMPLETED ASSEMBLY IN CLEAN, STATIC DISSIPATIVE PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35. STORE IN A PROJECT APPROVED STORAGE AREA.
11. △11 TRIM ITEM 31 CONNECTION WIRES TO LENGTH WHICH ALLOWS PROPER MOUNTING OF HOUSING TO BOARD AND INSPECTION OF CONNECTIONS.
12. REFERENCE SCHEMATIC: S1101EFS0300
13. EXPEDITION CONTROL FILES: S1101EFA0300_REV_B.ZIP

NOTES: UNLESS OTHERWISE SPECIFIED

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. DEMOISTURIZE (BAKE OUT) PWB (ITEM 1) FOR A MINIMUM OF 2 HOURS AT 93°C ±5.5°C (200°F ±10°F.) PERFORM SOLDERING IMMEDIATELY AFTER BAKING, OR STORE IN A CLEAN, DRY, SEALED STATIC DISSIPATIVE BAG OR CONTAINER.
5. △5 MARK THE ASSEMBLY WITH REV AND S/N USING ITEM 27.
6. ASSEMBLY, SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.2, NASA-STD-8739.3, AND THE FOLLOWING NOTES:
   - A. INSTALL AND SOLDER ALL SURFACE MOUNT (SMT) PART BEFORE THROUGH HOLE PARTS USING SOLDER (ITEM 30.)
   - B. FOR THROUGH HOLE PARTS, LEAD TERMINATIONS SHALL BE STRAIGHT THROUGH (NOT LAPPED OR CLINCHED.)
   - △C TIGHTEN FASTENERS SNUG TIGHT PER ESTABLISHED PROCEDURES.
   - D. DO NOT SUBMERGE ITEM 31 INTO CLEANING SOLVENT.
   - E. DISCARD WASHERS AND NUTS FROM ITEMS 7, 8 AND 9.
7. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY PRIOR TO CONFORMAL COATING PER TEST PLAN ACME-TST-026.

REV	DESCRIPTION	APPROVED
B	REVISE PER ACME-ECO-251 & ACME-ECO-329	TAG 10/14/14 DAS 10/8/14 RJP 10/8/14
A	REVISE PER ACME-ECO-140	TAG 7/30/13 DAS 7/31/13 RJP 7/30/13

REVISION HISTORY

S1101MFA0100
NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 10/14/14
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 3/22/13	QUALITY	D. CHOVAN 3/26/13
REVIEWER	M. HOMKO 3/25/13	MATERIALS	M. WILKINSON 3/22/13
MECHANICAL	A. DRAKE 3/22/13	PROJ LEAD	B. BOROWSKI 3/26/13
ELECTRICAL	T. GOBELI 3/22/13	-	-
STRUCTURAL	W. BORRELLI 3/22/13	-	-
SAFETY	C. WERNER 3/25/13	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
DELIVERY CODE: SPACE110

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CLEVELAND, OHIO

TITLE
ACME
DOWN STREAM RELAY BOARD
ASSEMBLY DRAWING

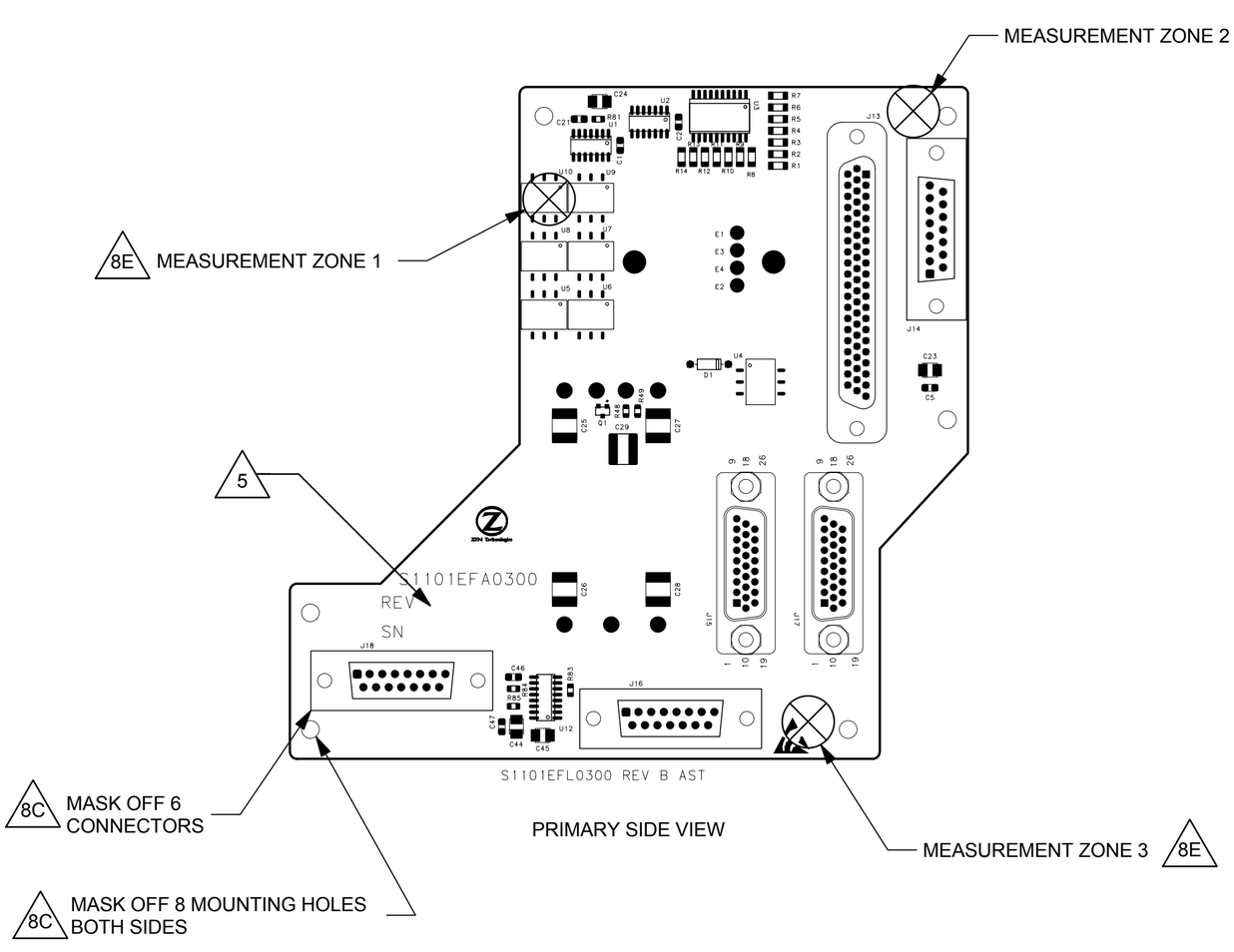
SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFA0300	B

SCALE: NONE | INITIAL REL DATE 3/26/13 | SHEET 1 OF 3

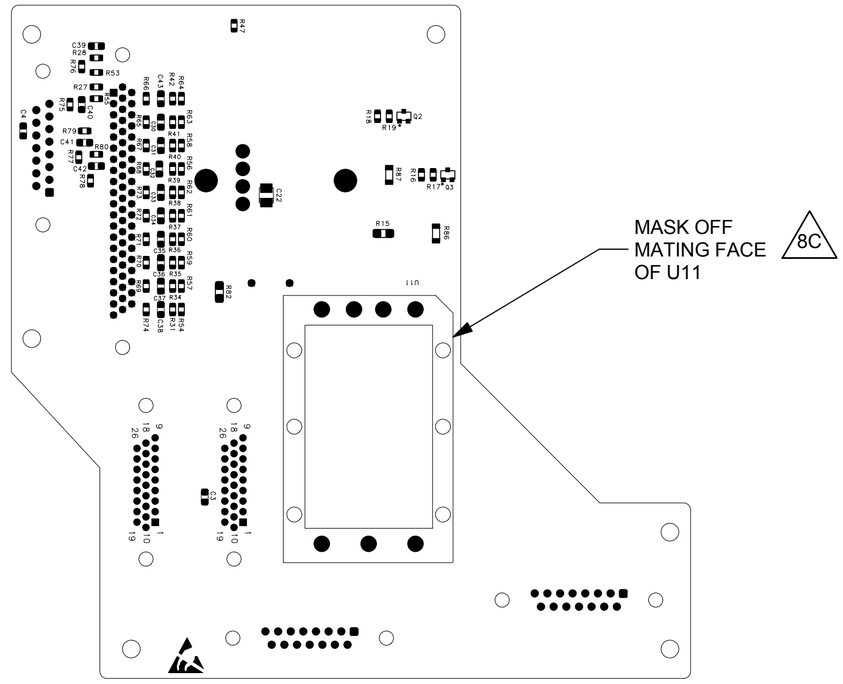
RELEASE LEVEL: .
MODEL NAME: S1101EFA0300_REV_B.bpd

BLUEPRINT
VERSION 3.0

REV B
SH 1
DWG NO S1101EFA0300



PRIMARY SIDE VIEW

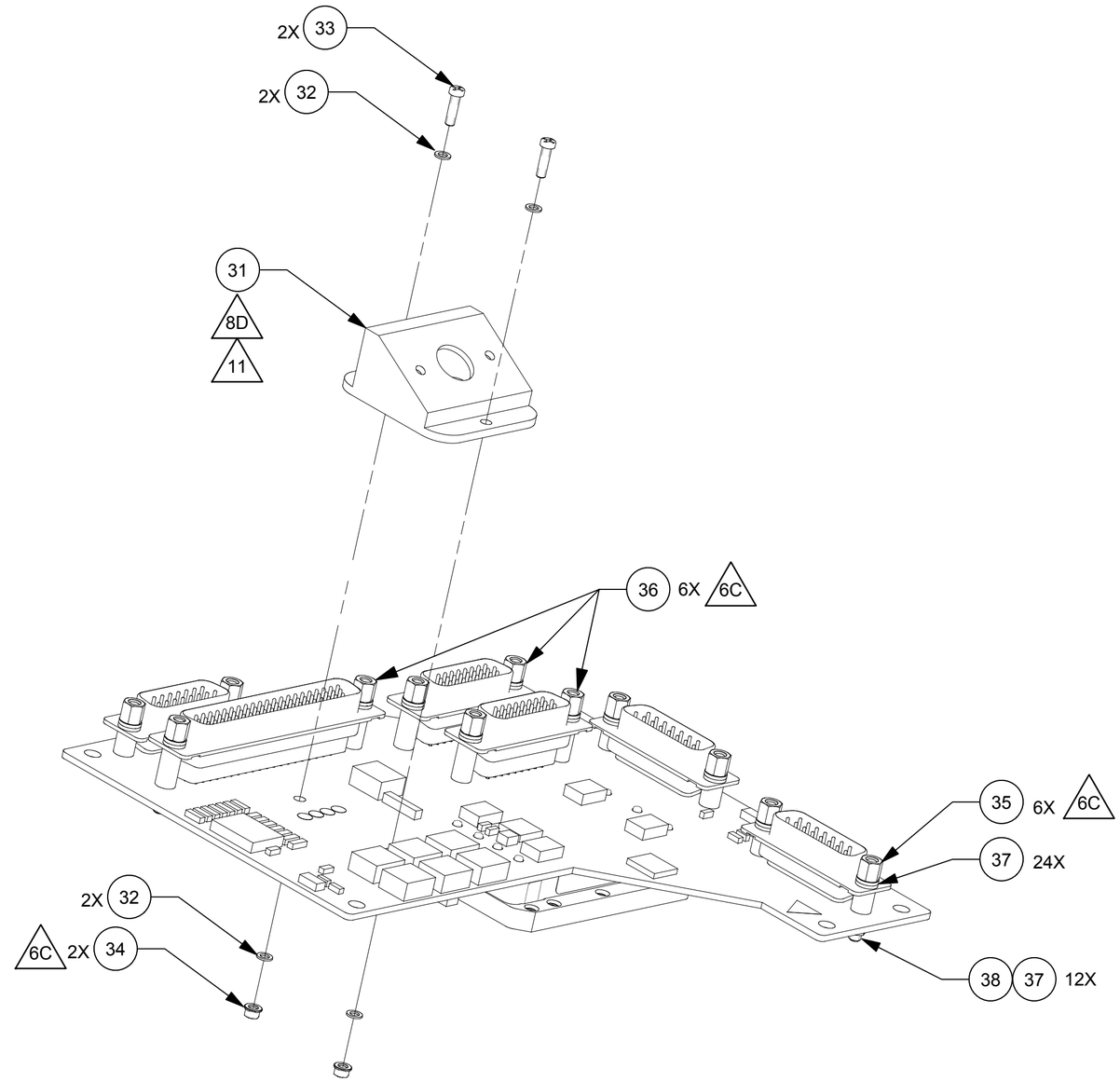


SECONDARY SIDE VIEW

SIZE	CAGE CODE	DWG NO	REV
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SCALE: NONE	INITIAL REL DATE 3/26/13	SHEET 2 OF 3	

BLUEPRINT
VERSION 3.0

REV B
SH 2
DWG NO S1101EFA0300



RADIOMETER TO RELAY BOARD CONNECTIONS	
RADIOMETER WIRES	RELAY BOARD E-POINTS
RED	E2
BLACK	E4
VIOLET	E3
ORANGE	E1

SIZE	CAGE CODE	DWG NO	REV
C	0SG68	S1101EFA0300	B
SCALE: NONE	INITIAL REL DATE 3/26/13	SHEET 3 OF 3	

BLUEPRINT
VERSION 3.0

DWG NO S1101EFA0300 SH 3 REV B